

Deliverable 1 – Project Understanding Report

Healthcare Patient Experience Analytics using Power BI

Organization Overview

The **National Healthcare Quality Commission (NHQC)** is a healthcare regulatory organization responsible for monitoring and improving patient experience across hospitals in the United States. The organization collects and analyzes patient feedback through the **Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS)** survey, which measures various aspects of healthcare quality, including communication with doctors and nurses, hospital cleanliness, pain management, discharge information, and overall hospital recommendation.

The primary objective of NHQC is to transform patient feedback into actionable insights that support healthcare providers, policy makers, and administrators in improving healthcare quality, patient satisfaction, and hospital performance nationwide.

Business Problem

Healthcare organizations collect millions of patient survey responses every year through the HCAHPS survey. Although the data contains valuable information regarding patient satisfaction and healthcare quality, it is distributed across multiple states, regions, reporting periods, and quality measures. This makes it difficult for healthcare administrators to monitor performance, identify improvement opportunities, compare results across different locations, and benchmark hospitals against national averages.

The absence of a centralized Business Intelligence solution limits timely decision-making and reduces the effectiveness of healthcare quality improvement initiatives. Therefore, an interactive Power BI dashboard is required to consolidate healthcare data, monitor key performance indicators (KPIs), analyze trends, and provide actionable insights for executive decision-making.

Project Objectives

The primary objective of this project is to develop an interactive Business Intelligence solution using Microsoft Power BI to analyze healthcare patient experience data and support data-driven decision-making.

The specific objectives are:

- Develop an interactive Power BI dashboard for healthcare performance monitoring.
 - Analyze patient satisfaction at national, regional, and state levels.
 - Evaluate healthcare quality measures using HCAHPS survey data.
 - Compare state performance against national benchmarks.
 - Identify high-performing and low-performing healthcare quality measures.
 - Monitor patient satisfaction trends across reporting periods.
 - Develop DAX measures for healthcare KPI calculations.
 - Design an optimized Star Schema data model for efficient reporting.
 - Generate actionable business insights to support healthcare quality improvement.
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Project Scope

This project focuses on analyzing historical HCAHPS survey data using Microsoft Power BI.

The scope includes:

- National patient satisfaction analysis
- State-wise performance analysis
- Regional performance comparison
- Patient experience measure analysis
- National versus state benchmarking
- Reporting period trend analysis
- KPI development using DAX
- Interactive dashboard development
- Star Schema data modeling
- Business insight generation

This project is limited to descriptive and diagnostic analytics and does not include predictive analytics or machine learning models.

Key Business Questions

The project aims to answer the following business questions:

1. Which states consistently deliver the best patient experience?
2. Which healthcare quality measures receive the highest patient satisfaction?
3. Which measures require immediate improvement?
4. How has patient satisfaction changed over different reporting periods?
5. Which U.S. regions consistently outperform others?
6. How do state-level results compare with national averages?
7. Which patient experience dimensions have the greatest impact on hospital recommendations?

8. Which regions and states should be prioritized for healthcare quality improvement initiatives?
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Assumptions

The following assumptions were considered during the project:

- The HCAHPS survey data is accurate and complete.
 - Survey responses accurately represent patient experiences.
 - Reporting periods are standardized across the dataset.
 - Percentage values are valid and comparable.
 - Relationships between datasets are correctly established.
 - Dashboard users will interact with reports using filters and slicers.
 - The solution is intended to support strategic and operational decision-making.
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Expected Outcomes

The successful completion of this project will provide:

- A centralized healthcare analytics dashboard.
 - Better visibility into national, regional, and state-level patient satisfaction.
 - Faster identification of high-performing and underperforming healthcare measures.
 - Effective benchmarking against national averages.
 - Improved monitoring of healthcare quality trends.
 - Interactive dashboards for executive reporting.
 - Actionable insights that support evidence-based healthcare decision-making.
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Stakeholders

Stakeholder	Responsibility
Chief Executive Officer (CEO)	Monitor overall healthcare performance
Chief Quality Officer (CQO)	Evaluate healthcare quality and patient satisfaction
Director of Patient Experience	Improve patient experience initiatives
State Health Administrators	Monitor state-level healthcare performance
Hospital Accreditation Team	Benchmark hospitals against quality standards
Policy Makers	Develop healthcare policies using analytical insights
Healthcare Analytics Team	Develop dashboards, KPIs, and business reports

Conclusion

This project delivers an enterprise-level Business Intelligence solution that enables healthcare stakeholders to analyze patient experience data through interactive dashboards, KPI monitoring, and trend analysis. By transforming healthcare survey data into meaningful insights, the solution supports evidence-based decision-making, improves healthcare quality, and helps organizations identify opportunities for continuous improvement in patient satisfaction.

